Lecture 17: Section 5.3

Introduction to u-Substitution

Md Ahsanul Haque, Ph.D.

Assistant Professor
Department of Mathematics and Physics
North South University, Dhaka

What is u-Substitution?

Definition (u-Substitution)

u-Substitution is an integration technique that helps us integrate composite functions by making a substitution that simplifies the integral.

The Big Idea

u-Substitution is the **reverse process** of the chain rule from differentiation.

- Chain Rule: $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$
- u-Substitution: $\int f'(g(x)) \cdot g'(x)dx = f(g(x)) + C$

Example

If we see $\int 2x \cdot e^{x^2} dx$, we notice:

- x^2 is the "inside function"
- $2x$ is the derivative of x^2
- Perfect candidate for u-substitution!

When to Use u-Substitution

Look for This Pattern

$$\int f(g(x)) \cdot g'(x)dx$$

Where:

- $g(x)$ is the "inside function"
- $g'(x)$ is the derivative of the inside function
- $f(g(x))$ is some function of the inside function

Key Indicators

- A composition of functions
- The derivative of the inner function appears as a factor
- The integrand looks like "something times derivative of that something"

Example

Good candidates: $\int 2x\cos(x^2)dx$, $\int 3x^2e^{x^3}dx$, $\int \frac{1}{x}\ln(x)dx$

Step-by-Step u-Substitution

The Procedure

1. **Choose u:** Identify the inner function and let $u = g(x)$
2. **Find du:** Differentiate: $du = g'(x)dx$
3. **Substitute:** Replace all x terms with u terms
4. **Integrate:** Evaluate the simpler integral in terms of u
5. **Back-substitute:** Replace u with $g(x)$
6. **Add C:** Don't forget the constant of integration!

Important Check

After step 2, make sure ALL x terms can be expressed in terms of u. If not, u-substitution might not work!

Example 1: Basic u-Substitution

Evaluate $\int 2x\cos(x^2)dx$

Solution:

1. **Choose u:** The inner function is x^2, so $u = x^2$
2. **Find du:** $du = 2xdx$
3. **Substitute:** $\int 2x\cos(x^2)dx = \int \cos(u)du$
4. **Integrate:** $\int \cos(u)du = \sin(u) + C$
5. **Back-substitute:** $\sin(u) + C = \sin(x^2) + C$

Verification:

$$\frac{d}{dx}[\sin(x^2) + C] = \cos(x^2) \cdot 2x = 2x\cos(x^2)$$

Example 2: Power Rule with u-Substitution

Evaluate $\int 3x^2(x^3+1)^4 dx$

Solution:

1. **Choose u:** The inner function is x^3+1, so $u = x^3+1$
2. **Find du:** $du = 3x^2 dx$
3. **Substitute:** $\int 3x^2(x^3+1)^4 dx = \int u^4 du$
4. **Integrate:** $\int u^4 du = \frac{u^5}{5} + C$
5. **Back-substitute:** $\frac{u^5}{5} + C = \frac{(x^3+1)^5}{5} + C$

Verification:

$$\frac{d}{dx}\left[\frac{(x^3+1)^5}{5} + C\right] = (x^3+1)^4 \cdot 3x^2 = 3x^2(x^3+1)^4$$

Example 3: When du Doesn't Match Exactly

Evaluate $\int x\sqrt{x^2+4}dx$

Solution:

1. **Choose u:** $u = x^2 + 4$
2. **Find du:** $du = 2xdx$
3. **Problem:** We have xdx but need $2xdx$
4. **Solve:** From $du = 2xdx$, we get $xdx = \frac{du}{2}$
5. **Substitute:** $\int x\sqrt{x^2+4}dx = \int \sqrt{u} \cdot \frac{du}{2} = \frac{1}{2}\int u^{1/2}du$
6. **Integrate:** $\frac{1}{2} \cdot \frac{u^{3/2}}{3/2} + C = \frac{1}{2} \cdot \frac{2}{3}u^{3/2} + C = \frac{1}{3}u^{3/2} + C$
7. **Back-substitute:** $\frac{1}{3}(x^2+4)^{3/2} + C$

Adjusting Constants

Key Technique

If du doesn't exactly match what you have, you can:

- Solve for the missing part: $dx = \frac{du}{g'(x)}$
- Adjust constants by multiplying/dividing
- Make sure ALL x terms disappear after substitution

Example

For $\int xe^{x^2} dx$:

- $u = x^2$, $du = 2xdx$
- We have xdx, but need $2xdx$
- So $xdx = \frac{du}{2}$
- Substitute: $\int e^u \cdot \frac{du}{2} = \frac{1}{2} \int e^u du$

Example 4: Trigonometric u-Substitution

Evaluate $\int \sin(x)\cos(x)dx$

Solution:

1. **Choose u:** Let $u = \sin(x)$ (could also use $u = \cos(x)$)
2. **Find du:** $du = \cos(x)dx$
3. **Substitute:** $\int \sin(x)\cos(x)dx = \int u du$
4. **Integrate:** $\int u du = \frac{u^2}{2} + C$
5. **Back-substitute:** $\frac{u^2}{2} + C = \frac{\sin^2(x)}{2} + C$

Alternative using $u = \cos(x)$:

1. $u = \cos(x)$, $du = -\sin(x)dx$
2. $\int \sin(x)\cos(x)dx = -\int u du = -\frac{u^2}{2} + C = -\frac{\cos^2(x)}{2} + C$

Note: Both answers are correct! They differ by a constant.

Example 5: Logarithmic u-Substitution

Evaluate $\int \frac{\ln(x)}{x} dx$

Solution:

1. **Choose u:** $u = \ln(x)$
2. **Find du:** $du = \frac{1}{x} dx$
3. **Substitute:** $\int \frac{\ln(x)}{x} dx = \int u du$
4. **Integrate:** $\int u du = \frac{u^2}{2} + C$
5. **Back-substitute:** $\frac{u^2}{2} + C = \frac{(\ln(x))^2}{2} + C$

Verification:

$$\frac{d}{dx}\left[\frac{(\ln(x))^2}{2} + C\right] = \ln(x) \cdot \frac{1}{x} = \frac{\ln(x)}{x}$$

Example 6: Definite Integrals with u-Substitution

Evaluate $\int_1^2 3x^2(x^3+1)^2 dx$

Solution:

1. **Choose u:** $u = x^3 + 1$
2. **Find du:** $du = 3x^2 dx$
3. **Change limits:**
   - When $x = 1$: $u = 1^3 + 1 = 2$
   - When $x = 2$: $u = 2^3 + 1 = 9$
4. **Substitute:** $\int_1^2 3x^2(x^3+1)^2 dx = \int_2^9 u^2 du$
5. **Integrate:** $\int_2^9 u^2 du = \left[\frac{u^3}{3}\right]_2^9$
6. **Evaluate:** $\frac{9^3}{3} - \frac{2^3}{3} = \frac{729}{3} - \frac{8}{3} = \frac{721}{3}$

Note: For definite integrals, you can either:

- Change the limits and evaluate in u-space, OR
- Back-substitute and use original limits

Recognizable Patterns

Common u-Substitution Forms

- $\int f(ax+b)dx \rightarrow u = ax+b$
- $\int x^n f(x^{n+1})dx \rightarrow u = x^{n+1}$
- $\int \frac{f'(x)}{f(x)}dx \rightarrow u = f(x)$
- $\int e^{f(x)}f'(x)dx \rightarrow u = f(x)$
- $\int \sin(f(x))f'(x)dx \rightarrow u = f(x)$
- $\int \cos(f(x))f'(x)dx \rightarrow u = f(x)$

The "Chain Rule Backwards" Pattern

Look for: [function of something] $\times$ [derivative of that something]

The $\frac{du}{u}$ Pattern

Very Common Pattern

$$\int \frac{f'(x)}{f(x)}dx = \ln|f(x)| + C$$

Example

- $\int \frac{2x}{x^2+1}dx = \ln|x^2+1| + C$
- $\int \frac{\cos(x)}{\sin(x)}dx = \ln|\sin(x)| + C$
- $\int \frac{e^x}{e^x+1}dx = \ln|e^x+1| + C$

Why it works

$u = f(x)$, $du = f'(x)dx$, so $\int \frac{f'(x)}{f(x)}dx = \int \frac{du}{u} = \ln|u| + C$

Trigonometric Patterns

Common Trigonometric u-Substitutions

- $\int \tan(x)dx = \int \frac{\sin(x)}{\cos(x)}dx \rightarrow u = \cos(x)$
- $\int \cot(x)dx = \int \frac{\cos(x)}{\sin(x)}dx \rightarrow u = \sin(x)$
- $\int \sec(x)dx \rightarrow$ tricky! (needs special technique)
- $\int \csc(x)dx \rightarrow$ tricky! (needs special technique)

Example

$\int \tan(x)dx$

- $u = \cos(x)$, $du = -\sin(x)dx$
- $\int \frac{\sin(x)}{\cos(x)}dx = -\int \frac{du}{u} = -\ln|u| + C = -\ln|\cos(x)| + C = \ln|\sec(x)| + C$

When u-Substitution Doesn't Work

Common Issues

- **Missing constant:** The derivative doesn't match exactly and can't be adjusted
- **Extra x terms:** After substitution, x terms remain
- **Wrong pattern:** The integrand doesn't fit the chain rule pattern

What to Try Instead

- Algebraic simplification
- Trigonometric identities
- Integration by parts (more advanced)
- Partial fractions (more advanced)

Example

$\int x\sqrt{x+1}dx \rightarrow$ u-substitution doesn't work well because derivative of $(x+1)$ is 1, but we have x not 1.

Practice Problems - Basic u-Substitution (8 problems)

Evaluate each integral using u-substitution:

1. $\int 2x(x^2+3)^4 dx$
2. $\int 3x^2 e^{x^3} dx$
3. $\int \frac{2x}{x^2+1} dx$
4. $\int \cos(x)\sin(x) dx$
5. $\int 4x^3(x^4-2)^5 dx$
6. $\int \frac{\ln(x)}{x} dx$
7. $\int x\sqrt{x^2+9} dx$
8. $\int \frac{e^x}{e^x+2} dx$

Practice Problems - Advanced u-Substitution (10 problems)

Evaluate each integral:

1. $\int x(x^2+1)^3 dx$
2. $\int \frac{x}{\sqrt{x^2+4}} dx$
3. $\int \sin(3x) dx$
4. $\int x^2(2x^3+5)^4 dx$
5. $\int \frac{\cos(x)}{\sin^2(x)} dx$
6. $\int e^{2x} dx$
7. $\int \frac{2x+1}{x^2+x+1} dx$
8. $\int \tan(x) dx$
9. $\int x\sqrt[3]{x^2-1} dx$
10. $\int \frac{\arctan(x)}{1+x^2} dx$

Practice Problems - Definite Integrals (8 problems)

Evaluate each definite integral:

1. $\int_0^1 2x(x^2+1)^3 dx$
2. $\int_1^2 \frac{3x^2}{x^3+1} dx$
3. $\int_0^{\pi/2} \sin(x)\cos(x) dx$
4. $\int_0^1 xe^{x^2} dx$
5. $\int_1^4 \frac{1}{\sqrt{x}(1+\sqrt{x})^2} dx$
6. $\int_0^{\pi/4} \sec^2(x)\tan(x) dx$
7. $\int_1^2 \frac{\ln(x)}{x} dx$
8. $\int_0^1 \frac{x}{x^2+1} dx$

Challenge Problems

1. Prove that $\int \frac{f'(x)}{f(x)}dx = \ln|f(x)| + C$ using u-substitution
2. Show that $\int \sec(x)dx = \ln|\sec(x) + \tan(x)| + C$ using an appropriate substitution
3. Find $\int \frac{1}{x\ln(x)}dx$
4. Evaluate $\int \frac{e^x - e^{-x}}{e^x + e^{-x}}dx$
5. Determine $\int x\sqrt{1-x^2}dx$ using two different u-substitutions

Less Apparent Substitutions

Sometimes the substitution isn't immediately obvious. We need to:

1. **Rewrite** the integrand algebraically
2. **Use identities** (trigonometric, logarithmic, etc.)
3. **Choose** u strategically after rewriting
4. **Adjust** constants as needed

Key Insight

These problems look harder at first, but they follow the same principle: **Find** $f(g(x)) \cdot g'(x)$ pattern, even if it's hidden!

Example

Consider $\int x^2\sqrt{x-1}\,dx$. The $\sqrt{x-1}$ suggests $u = x - 1$, but then we have x^2 which isn't the derivative.

Example 7: Algebraic Manipulation First

Evaluate $\int x^2\sqrt{x-1}\,dx$

Solution:

1. **Let** $u = x - 1$, so $x = u + 1$ and $dx = du$
2. **Rewrite** x^2: $x^2 = (u+1)^2 = u^2 + 2u + 1$

$$\begin{aligned}
\int x^2\sqrt{x-1}\,dx &= \int (u^2+2u+1)\sqrt{u}\,du \\
&= \int (u^2+2u+1)u^{1/2}\,du \\
&= \int (u^{5/2}+2u^{3/2}+u^{1/2})\,du \\
&= \frac{u^{7/2}}{7/2} + 2\cdot\frac{u^{5/2}}{5/2} + \frac{u^{3/2}}{3/2} + C \\
&= \frac{2}{7}u^{7/2} + \frac{4}{5}u^{5/2} + \frac{2}{3}u^{3/2} + C \\
&= \frac{2}{7}(x-1)^{7/2} + \frac{4}{5}(x-1)^{5/2} + \frac{2}{3}(x-1)^{3/2} + C
\end{aligned}$$

Example 8: Trigonometric Identities First

Evaluate $\int \cos^3(x)\,dx$

Solution:

1. **Rewrite using identity**: $\cos^2(x) = 1 - \sin^2(x)$

$$\cos^3(x) = \cos^2(x)\cos(x) = (1 - \sin^2(x))\cos(x)$$

2. **Now see the pattern!** We have a function of $\sin(x)$ times $\cos(x)$
3. **Let** $u = \sin(x)$, then $du = \cos(x)\,dx$
4. **Substitute**:

$$\int \cos^3(x)\,dx = \int (1 - u^2)\,du$$

5. **Integrate**:

$$= u - \frac{u^3}{3} + C$$

6. **Back-substitute** $u = \sin(x)$:

$$= \sin(x) - \frac{\sin^3(x)}{3} + C$$

Why That Trigonometric Substitution Worked

The Key Observation

For integrals involving odd powers of sine or cosine:

- $\int \sin^n(x)\cos^m(x)\,dx$
- If n is odd, save one $\sin(x)$ and convert the rest to $\cos(x)$ using $\sin^2(x) = 1 - \cos^2(x)$, then let $u = \cos(x)$
- If m is odd, save one $\cos(x)$ and convert the rest to $\sin(x)$ using $\cos^2(x) = 1 - \sin^2(x)$, then let $u = \sin(x)$

Example

Try $\int \sin^5(x)\,dx$:

- $\sin^5(x) = \sin^4(x)\sin(x) = (1 - \cos^2(x))^2 \sin(x)$
- Let $u = \cos(x)$, $du = -\sin(x)\,dx$
- Then $\int (1-u^2)^2(-du) = -\int (1 - 2u^2 + u^4)\,du$

Integrals Yielding Inverse Trigonometric Functions

The Six Fundamental Inverse Trig Integrals ($a > 0$)

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin\left(\frac{x}{a}\right) + C \qquad \int \frac{-dx}{\sqrt{a^2 - x^2}} = \arccos\left(\frac{x}{a}\right) + C$$

$$\int \frac{dx}{a^2 + x^2} = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C \qquad \int \frac{-dx}{a^2 + x^2} = \frac{1}{a}\operatorname{arccot}\left(\frac{x}{a}\right) + C$$

$$\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a}\operatorname{arcsec}\left(\frac{|x|}{a}\right) + C \qquad \int \frac{-dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a}\operatorname{arccsc}\left(\frac{|x|}{a}\right) + C$$

Example 9: Inverse Trigonometric Patterns

Evaluate $\int \frac{x}{\sqrt{1-x^4}}\,dx$

Solution:

1. **Notice the pattern**: This looks like the derivative of $\arcsin(x^2)$ or something similar
2. **Rewrite**: $\int \frac{x}{\sqrt{1-(x^2)^2}}\,dx$
3. **Let** $u = x^2$, then $du = 2x\,dx$, so $x\,dx = \frac{du}{2}$
4. **Substitute**:
$$\int \frac{x}{\sqrt{1-x^4}}\,dx = \int \frac{1}{\sqrt{1-u^2}} \cdot \frac{du}{2} = \frac{1}{2}\int \frac{du}{\sqrt{1-u^2}}$$
5. **Recognize the integral**: $\int \frac{du}{\sqrt{1-u^2}} = \arcsin(u) + C$
6. **So**:
$$\frac{1}{2}\int \frac{du}{\sqrt{1-u^2}} = \frac{1}{2}\arcsin(u) + C$$
7. **Back-substitute** $u = x^2$:
$$= \frac{1}{2}\arcsin(x^2) + C$$

Example 10: Another Inverse Trig Pattern

Evaluate $\int \frac{e^x}{1+e^{2x}}\,dx$

Solution:

1. **Rewrite denominator**: $1 + e^{2x} = 1 + (e^x)^2$
2. **Now it looks like**: $\int \frac{\text{something}}{1+(\text{something})^2}\,dx$
3. **Let** $u = e^x$, then $du = e^x\,dx$
4. **Perfect!** The numerator is exactly $e^x\,dx = du$
5. **Substitute**:
$$\int \frac{e^x}{1+e^{2x}}\,dx = \int \frac{du}{1+u^2}$$
6. **Recognize the integral**: $\int \frac{du}{1+u^2} = \arctan(u) + C$
7. **Back-substitute** $u = e^x$:
$$= \arctan(e^x) + C$$

Practice: Less Apparent Substitutions (8 problems)

Evaluate each integral:

1. $\int x\sqrt{x+2}\,dx$
2. $\int \frac{x}{\sqrt{x+3}}\,dx$
3. $\int \sin^3(x)\,dx$
4. $\int \cos^5(x)\,dx$
5. $\int \frac{x^3}{\sqrt{1-x^4}}\,dx$
6. $\int \frac{1}{x\sqrt{1-(\ln x)^2}}\,dx$
7. $\int \frac{\arctan(x)}{1+x^2}\,dx$
8. $\int \frac{e^{2x}}{1+e^{4x}}\,dx$

Practice: More Challenging Substitutions (8 problems)

Evaluate each integral:

1. $\int \frac{\sqrt{\ln x}}{x}\,dx$
2. $\int x(x-1)^{10}\,dx$
3. $\int \frac{\sin(x)}{\cos^3(x)}\,dx$
4. $\int \frac{x}{1+x^4}\,dx$
5. $\int \frac{1}{x\sqrt{4-(\ln x)^2}}\,dx$
6. $\int \frac{\cos(x)}{\sin^2(x)+4\sin(x)+3}\,dx$
7. $\int \frac{e^x}{\sqrt{1-e^{2x}}}\,dx$
8. $\int \frac{x^2}{(x-1)^5}\,dx$

Tips for Less Apparent Substitutions

When to Try Algebraic Manipulation

- You see a radical like $\sqrt{ax+b}$ or $\sqrt[3]{ax+b}$
- Let $u =$ the inside of the radical
- Solve for x in terms of u
- Replace ALL x terms with u terms

When to Try Trigonometric Identities

- You see odd powers of $\sin(x)$ or $\cos(x)$
- Use $\sin^2(x) + \cos^2(x) = 1$ to convert
- Save one factor to become part of du
- Convert the rest using the identity

When to Look for Inverse Trig Patterns

- Forms like $\frac{1}{\sqrt{a^2-u^2}}$ or $\frac{1}{a^2+u^2}$
- The derivative of u appears in the numerator
- Complete the square if necessary
- Match to derivative formulas for arcsin, arctan, etc.

Summary: Spotting Hidden Substitutions

1. **Radical with linear inside**: $u =$ inside, solve for x
2. **Odd powers of trig functions**: Save one, use identity, $u =$ the other trig function
3. **Inverse trig patterns**: Look for $\frac{1}{\sqrt{1-u^2}}$ or $\frac{1}{1+u^2}$ forms
4. **Exponential forms**: Look for e^{ax} patterns, $u = e^{ax}$
5. **Logarithmic forms**: Look for $\frac{1}{x}$ with functions of $\ln x$

Final Check

After choosing u, make sure:

- ALL x terms can be written in terms of u
- The dx converts completely to du
- No leftover x terms remain after substitution

Summary of Key u-Substitution Rules

If $u = g(x)$, then:

$$\int f(g(x)) \cdot g'(x)dx = \int f(u)du$$

Step-by-Step Method

1. Choose $u = g(x)$ (the inner function)
2. Find $du = g'(x)dx$
3. Substitute ALL x terms with u terms
4. Integrate with respect to u
5. Back-substitute $u = g(x)$
6. Add $+C$

Either:

- Change limits: when $x = a$, $u = g(a)$; when $x = b$, $u = g(b)$
- Or back-substitute and use original limits

Common u-Substitution Patterns to Recognize

- $\int [\text{function of } u] \cdot [\text{derivative of } u] dx \rightarrow u = \text{inner function}$
- $\int \frac{du}{u} \rightarrow \ln|u| + C$
- $\int e^u du \rightarrow e^u + C$
- $\int \sin(u) du \rightarrow -\cos(u) + C$
- $\int \cos(u) du \rightarrow \sin(u) + C$
- $\int u^n du \rightarrow \frac{u^{n+1}}{n+1} + C \ (n \neq -1)$

Final Advice

- Practice recognizing the patterns
- Always verify your answer by differentiating
- If stuck, try a different choice for u
- Remember: u-substitution is the reverse chain rule!